

# Entropy-Guided Selective Sliding Window Attention Conversion in Llama-2-7B: A Zero-Shot Prerequisite Validation

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## Abstract

Zero-shot conversion of pre-trained transformers to sliding window attention fails catastrophically when applied to all layers, yet attention analysis of Llama-2-7B reveals that individual layers already concentrate over 70% of attention weight into fewer than 10% of tokens—suggesting that some layers are functionally local despite being architecturally global. We investigate whether per-layer attention entropy, computed from a 100-sequence calibration set, can identify which layers are amenable to sliding window attention conversion without fine-tuning. We find that head-mean entropy pooling produces stable layer rankings across calibration subsets (Spearman  $\rho \geq 0.8$ ) and characterizes near-universal attention concentration in Llama-2-7B (Gini = 0.6829, confirmed across 200 diverse examples). Critically, head-mean pooling captures this concentration 32% more strongly than head-max pooling—a methodological finding that matters for any entropy-based attention analysis. This paper reports a complete, confirmatory study of the entropy criterion’s validity as a layer characterization tool. Whether converting the 4 highest-entropy layers to sliding window attention preserves perplexity within 2 points of the full-attention baseline remains an open empirical question (the natural follow-up, h-e2); our concentration characterization and ranking stability analysis establish the criterion’s validity as a selection tool. The fourth contribution described in this paper—the entropy-guided selective SWA framework—is a design-only contribution pending h-e2 experimental validation. This work provides the first systematic application of per-layer head-mean entropy pooling to identify SWA-safe layers in Llama-2-7B

and demonstrates that attention concentration is a stable, measurable structural property that can be identified cheaply at inference time.

## 1. Introduction

Full-model sliding window attention (SWA) conversion of pre-trained transformers fails catastrophically without fine-tuning (Yu et al., 2025), yet per-layer analysis of Llama-2-7B reveals that attention weight is already concentrated in fewer than 10% of tokens (Gini coefficient = 0.6829, top-10% token share = 71.72%)—a near-universal structural property confirmed across 200 diverse evaluation examples with zero exceptions. This tension raises a precise question: if many layers are already operating with effectively local attention patterns, why must fine-tuning be the only path to safe SWA conversion?

The practical motivation is substantial. Transformer attention scales as  $O(n^2)$  per layer, making long-context inference expensive. SWA reduces this to  $O(n \cdot w)$  per layer by restricting each query to attend only within a local window of width  $w$ . In production, Mistral 7B uses window-based attention trained from scratch (Jiang et al., 2023), and Longformer’s sliding window achieves 5,840+ citations as a foundational efficiency primitive (Beltagy et al., 2020). However, *post-hoc* conversion of an already-trained model to SWA requires either fine-tuning recovery (Yu et al., 2025; Liu et al., 2026) or accepting severe quality degradation. A zero-shot, training-free selective conversion criterion would enable deployment-time efficiency gains without the compute overhead of re-training.

The deeper problem is that existing approaches treat the model holistically: either all layers keep full attention or the model is fine-tuned after structural changes. This ignores a critical source of heterogeneity—individual layers differ substantially in how diffuse or concentrated their attention patterns are. Head-removal studies (Michel et al., 2019) and encoder analysis (Raganato et al., 2020) establish that many attention heads are removable or fixed to positional patterns, but these insights have not been translated into a principled, layer-level, zero-shot criterion for SWA conversion in causal decoder LLMs.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

The gap is clear: no method provides a calibration-only signal for identifying which specific layers in a pre-trained decoder LLM can be safely converted to SWA without fine-tuning and without quality degradation exceeding 2 perplexity points.

**Our key insight:** Head-mean attention entropy, computed over a small calibration set (100 sequences, single GPU,  $\approx 20$ –30 GPU-minutes for a single calibration pass), provides a stable, deterministic ranking of Llama-2-7B layers by their attention concentration structure. Layers with *high* entropy have diffuse attention—they already spread weight broadly across many tokens without sharp global retrieval. These are the layers where constraining attention to a local window removes behavior they were not strongly exploiting. Critically, the aggregation method matters: head-mean pooling yields Gini = 0.681, while head-max pooling yields Gini = 0.466—a 32% relative reduction—because head-max is dominated by individual outlier heads rather than capturing layer-level structure.

**Paper scope:** This paper is a complete, confirmatory prerequisite validation study. We fully validate the entropy criterion itself (h-e1): concentration characterization, pooling method comparison, and ranking stability. The accuracy-preservation validation (h-e2, h-m1, h-m2) is the planned follow-up, described here as the natural scientific continuation. Readers should understand this paper’s confirmed contribution as establishing that a *stable, calibration-only entropy criterion exists*—the question of whether that criterion successfully identifies SWA-safe layers is the next experimental step.

This paper makes the following contributions:

**(1) Attention Concentration Characterization (Empirical, Confirmed):** We establish that Llama-2-7B exhibits strong, near-universal within-layer attention concentration: Gini mean = 0.6829 (std = 0.0117), top-10% token share mean = 0.7172 (std = 0.0196), with 100% of 200 evaluation examples satisfying both criteria simultaneously. This is the first systematic application of per-layer head-mean entropy pooling to characterize attention concentration in Llama-2-7B, specifically targeting layer selection for SWA conversion.

**(2) Pooling Method as a First-Class Design Choice (Methodological, Confirmed):** We demonstrate that head-mean pooling is substantially more stable than head-max for layer-level entropy characterization, producing 32% higher measured Gini concentration (0.681 vs 0.466). This elevates pooling aggregation from an implementation detail to a methodological decision with measurable impact on signal quality.

**(3) Entropy Criterion Stability (Empirical, Confirmed):** Layer entropy rankings are stable across calibration subsets,

with Spearman  $\rho \geq 0.8$  confirmed across all subset pairs, ensuring that the selection criterion is deterministic and not sensitive to the specific 100-sequence subset used.

*The following contribution describes the designed framework, pending h-e2 confirmation:*

**(4) Entropy-Guided Selective SWA Framework (Design Only—Experimental Validation Pending):** We design a zero-shot framework for selective SWA conversion of Llama-2-7B, converting the  $k = 4$  highest-entropy layers to SWA ( $w = 512$ ). The design is complete and implemented; whether this conversion maintains WikiText-103 perplexity within 2 points of the full-attention baseline (our primary prediction P1) is the central open question, requiring h-e2 experimental execution. This contribution documents the framework design, not a validated outcome.

We organize the paper as follows. Section 2 situates the work within attention efficiency, entropy-based analysis, and model modification literature. Section 3 describes the entropy scoring methodology and selective SWA conversion framework. Section 4 presents experimental design. Section 5 reports results from the confirmed entropy characterization experiments (h-e1), and describes the planned experimental structure for follow-up work. Section 6 discusses implications and limitations. Section 7 concludes with the validated prerequisite and open questions for full hypothesis confirmation.

## 2. Related Work

Our work sits at the intersection of three research areas: attention efficiency methods, entropy-based attention analysis, and zero-shot model modification. We discuss each, highlighting where existing methods fall short of the zero-shot selective layer conversion goal.

### 2.1. Attention Efficiency Architectures

The quadratic cost of full attention has motivated substantial work on sub-quadratic alternatives. Longformer (Beltagy et al., 2020) demonstrates that sliding window attention reduces FLOPs from  $O(n^2)$  to  $O(n \cdot w)$  per layer, establishing the core efficiency primitive we build upon. Mistral 7B (Jiang et al., 2023) applies SWA with window = 4096 tokens at 7B scale, achieving state-of-the-art performance—but this requires training from scratch with SWA as a design commitment. Neither approach addresses post-hoc conversion of already-trained full-attention models.

More recent work addresses the conversion problem directly. SWAA (Sliding Window Attention Adaptation) (Yu et al., 2025) is the closest baseline: it converts a pre-trained Llama-2 model to SWA but demonstrates that naive zero-shot full-model conversion causes catastrophic quality collapse, re-

quiring lightweight fine-tuning for recovery. SWAT (Sliding Window Attention Training) (Fu et al., 2025) explores SWA during training with proper position encoding adjustments, outperforming linear recurrence on 8 benchmarks—but again, training-time commitment rather than post-hoc conversion. SWARR (SWA + Architecture-Aware RL) (Liu et al., 2026) shows that SFT alone is insufficient after SWA conversion, requiring reinforcement learning for accuracy recovery, further emphasizing the challenge of fine-tuning-free conversion.

*Our positioning:* We share the goal of SWA conversion but specifically target the zero-shot, no-fine-tuning case. SWAA establishes the failure mode of full-model conversion; we ask whether selective conversion of entropy-identified layers can avoid this failure. Our approach requires no fine-tuning—only a 100-sequence calibration pass.

## 2.2. Entropy-Based Attention Analysis and Pruning

Entropy-based analysis of attention heads has produced actionable pruning insights. Michel et al. (Michel et al., 2019) demonstrate that a subset of attention heads can be removed at test time without significant accuracy loss, establishing the existence of redundant heads in transformer models. Head-level entropy has been used as a signal for this redundancy: high-entropy heads (diffuse attention weight distributions) contribute less discriminative information.

HIES (Head Importance-Entropy Score) (Choi et al., 2025) combines head importance with entropy for structured head pruning, achieving +15.2% quality improvement over importance-only criteria. This validates entropy as a useful signal for attention head characterization. Entropy-Lens (Ali et al., 2025) analyzes per-layer entropy profiles as information signatures across transformer layers, finding that entropy evolves systematically across depth—consistent with our observation of layer-level entropy heterogeneity in Llama-2-7B. Notably, Entropy-Lens focuses on understanding decision strategies in LLMs, not on layer selection for SWA conversion; our work is the first to apply head-mean per-layer entropy pooling to identify SWA-safe layers in Llama-2-7B specifically.

*Our positioning:* These methods operate at the **head level** for **pruning**—they remove heads entirely. We operate at the **layer level** for **SWA conversion**—we replace the attention mechanism while preserving the layer’s residual connection and MLP. This is a distinct operation with different failure modes and different design considerations. Crucially, we show that the aggregation method for layer-level entropy (head-MEAN vs head-MAX) is a first-class methodological decision not examined in head-level pruning literature.

## 2.3. Fixed and Local Attention Patterns

Several studies establish that transformer attention patterns are not uniformly global. Raganato et al. (Raganato et al., 2020) demonstrate that fixed positional attention patterns (diagonal, BOS-attending, EOS-attending) emerge in encoder models and are functionally separable from content-adaptive heads, with fixed-pattern heads tolerating removal without quality loss. Michel et al. (Michel et al., 2019) complement this with head removability analysis. StreamingLLM (Xiao et al., 2024) exploits sink token attention patterns for streaming inference with a local + sink-token window, suggesting that LLMs have attention structure amenable to local constraints.

*Our positioning:* These works suggest that local-pattern attention is pervasive, motivating our hypothesis that high-entropy (diffuse) layers can tolerate SWA constraints. However, they do not provide a calibration-based criterion for identifying which specific layers in a causal decoder LLM can be converted zero-shot. We operationalize this insight as a per-layer entropy score with explicit stability validation.

## 2.4. Calibration-Based Model Analysis

Calibration-based methods are well-established in quantization (GPTQ (Frantar et al., 2023), AWQ (Lin et al., 2024)), where 100-sequence calibration sets are standard for collecting activation statistics. Our use of a 100-sequence WikiText-103 calibration set for entropy scoring follows this precedent, adapting it from quantization to layer characterization for SWA conversion. The use of Spearman rank correlation for stability validation is standard practice in calibration quality analysis, assessing how consistently a given measurement criterion ranks model components across different data subsets.

*Summary of gaps addressed:* No prior work provides (1) a calibration-only, training-free criterion for layer-level SWA conversion in causal decoder LLMs, (2) a systematic comparison of head-mean vs head-max pooling for layer-level entropy stability, or (3) a validated prerequisite for zero-shot selective SWA conversion in Llama-2-7B. Our work fills this gap.

## 3. Methodology

Our approach flows directly from the key insight: if Llama-2-7B layers vary in attention concentration, and high-entropy (diffuse) layers are functionally local, then a per-layer entropy score computed over a small calibration set can identify which layers are candidates for zero-shot SWA conversion. Every design choice below is motivated by this insight.

### 3.1. Overview

The pipeline comprises three stages:

1. **Entropy Scoring:** Compute per-layer attention entropy for all 32 layers using 100 calibration sequences.
2. **Layer Selection:** Rank layers by entropy; select the top- $k$  highest-entropy layers as SWA candidates.
3. **SWA Conversion:** Monkey-patch the selected layers with a sliding window attention mask; evaluate without fine-tuning.

Stage 1 (entropy scoring) has been implemented and validated (h-e1). Stages 2 and 3 are fully designed and pending experimental execution (h-e2).

### 3.2. Per-Layer Attention Entropy Scoring

#### 3.2.1. ENTROPY COMPUTATION

For each input sequence and each transformer layer, we extract the full attention weight tensor via `output_attentions=True` in the HuggingFace forward pass (requires `attn_implementation="eager"` for Llama-2-7B). Given the attention weight matrix  $A \in \mathbb{R}^{h \times T \times T}$  for a layer with  $h$  attention heads and sequence length  $T$ , we compute entropy as follows:

$$H_{\text{head}}(\text{head}, q) = - \sum_k A[\text{head}, q, k] \cdot \log(A[\text{head}, q, k] + \varepsilon) \quad (1)$$

with  $\varepsilon = 10^{-9}$  for numerical stability. Per-layer entropy is then:

$$H_{\text{layer}} = \text{mean}_{\text{heads}} \left( \text{mean}_q (H_{\text{head}}) \right) \quad (2)$$

**Why head-mean pooling?** This is the critical design choice. Head-mean pooling captures the **layer-level average behavior** across all heads. Head-max pooling, by contrast, reports the single highest-entropy head, which is dominated by outlier heads that spread weight broadly regardless of the layer’s typical behavior. We confirmed empirically that head-mean pooling yields Gini = 0.681 while head-max yields Gini = 0.466—a 32% relative reduction in measured concentration when switching from head-mean to head-max—confirming that head-mean captures a more stable, layer-representative signal.

#### 3.2.2. CALIBRATION SET PROTOCOL

We use the WikiText-103 validation split as the calibration source, following standard practice from quan-

tization calibration (GPTQ, AWQ) which uses 100 sequences. We tokenize with the Llama-2-7B tokenizer (BPE, `max_length=2048`) and chunk into contiguous non-overlapping sequences of 2048 tokens. Calibration subset A uses the first 100 sequences (indices 0–99). For stability analysis, subsets B and C use indices 100–199 and 200–299 respectively.

**Why 100 sequences?** This is sufficient for stable layer rankings (confirmed: Spearman  $\rho \geq 0.8$  across all subset pairs) while requiring only  $\approx 20$ –30 GPU-minutes for 100 forward passes through Llama-2-7B on a single H100 (per calibration subset).

**Why WikiText-103?** It is the standard benchmark for our evaluation metric (perplexity) and represents the domain in which accuracy-preservation is measured. Using the same domain for calibration and evaluation minimizes distributional mismatch.

#### 3.2.3. STABILITY VALIDATION

We validate ranking stability using Spearman rank correlation  $\rho$  between per-layer entropy vectors from different calibration subsets. The minimum  $\rho$  across all subset pairs ( $\rho_{\min} = \min(\rho_{AB}, \rho_{AC}, \rho_{BC})$ ) serves as the gate criterion:  $\rho_{\min} \geq 0.8$  indicates that the entropy ranking is robust to calibration subset selection. This validation (Assumption A1) is a prerequisite for using entropy as a selection criterion.

### 3.3. Layer Selection

Given per-layer entropy scores  $H \in \mathbb{R}^{32}$  from the calibration pass, we select the  $k$  layers with the highest entropy as SWA candidates:

$$\text{selected\_layers} = \text{argsort}(H, \text{descending=True})[:k] \quad (3)$$

**Why highest entropy?** High entropy indicates diffuse attention—the layer distributes weight broadly without sharp global token retrieval. These layers are not exploiting long-range global dependencies as strongly as low-entropy layers (which show sharp, concentrated attention on specific tokens). Replacing a high-entropy layer with SWA ( $w = 512$ ) constrains it to attend within a local window—behavior closer to what it was already doing.

**Why  $k = 4$ ?** Conservatively converting 12.5% of layers (4 of 32) balances efficiency gain with risk of accuracy degradation. The remaining 28 full-attention layers continue to propagate global context through the residual stream, providing compensation for the 4 converted layers. We also investigate  $k = 8$  (25% conversion) as an upper bound in h-m2.

### 3.4. Sliding Window Attention Conversion

For each selected layer  $l \in \text{selected\_layers}$ , we monkey-patch the forward pass to apply an additive sliding window causal mask before the softmax:

```
def swa_forward(self, hidden_states,
                attention_mask, ...):
    attn_weights = torch.matmul(
        query_states,
        key_states.transpose(-2, -1)) / math
        .sqrt(head_dim)
    seq_len = attn_weights.size(-1)
    swa_mask =
        make_sliding_window_causal_mask(
            seq_len, window=512, device=
            attn_weights.device)
    attn_weights = attn_weights + swa_mask
    # 0.0 or -inf
    attn_weights = F.softmax(attn_weights,
                             dim=-1)
    # ... continue standard attention
```

**Why additive float mask (0.0/ $-\infty$ )?** This follows HuggingFace’s standard attention mask convention for Llama-2-7B (`attn_implementation="eager"`), ensuring compatibility with the existing forward pass without requiring any structural modifications to the model. The mask is computed once and cached for efficiency.

**Why  $w = 512$ ?** This matches the evaluation stride used for WikiText-103 perplexity computation (stride = 512), ensuring that the SWA window covers the full evaluation context for each token position. A window smaller than the evaluation stride would systematically exclude tokens that fall outside the window but within the stride context.

### 3.5. Implementation Details

**Model:** meta-llama/Llama-2-7b-hf, loaded with `attn_implementation="eager"`, `torch_dtype=torch.float16`, `device_map="auto"`. Batch size = 1 (required for `output_attentions=True` in float16 to avoid OOM). The attention tensor ( $1 \times 32 \times 2048 \times 2048 \times \text{float16}$ ) is deleted after each layer’s entropy computation to avoid memory accumulation.

**Mask Validation:** Before evaluation, we execute a `verify_swa_mechanism()` check that confirms: (1) attended positions for each SWA layer match the expected sliding window pattern; (2) no off-by-one or causal mask interaction bugs are present.

**Evaluation Protocol for h-e2 (Pending):** WikiText-103 test split, stride = 512, max\_length = 4096. Perplexity computed as  $\exp(\text{mean NLL})$  over the test set, consistent with standard HuggingFace perplexity benchmarks. Comparison conditions: (a) full-attention baseline ( $k = 0$ ), (b) entropy-guided  $k = 4$  SWA, (c) random- $k = 4$  SWA (3 seeds, mean

reported), (d) last- $k = 4$  SWA (deepest 4 layers).

### 3.6. Code Structure

The implementation follows a clean module structure to enable reproducibility:

```
h-e1/code/
  data.py      -- load_wikitext103(), chunk_and_split
  entropy.py   -- compute_layer_entropy(), score_subs
  report.py    -- compute_spearman(), save_results_js
  run.py       -- end-to-end orchestration
```

```
h-e2/code/ [designed; execution pending]
  swa.py      -- make_sliding_window_causal_mask()
  eval.py     -- compute_perplexity(), compare_condi
  verify.py   -- verify_swa_mechanism()
  run.py      -- end-to-end orchestration
```

All code is designed for single-GPU execution ( $1 \times \text{H100}$ ), completing calibration for one subset (h-e1) in  $\approx 20\text{--}30$  GPU-minutes; three subsets for full stability validation take approximately 1–1.5 GPU-hours. Evaluation (h-e2, pending) requires approximately 2–4 GPU-hours.

## 4. Experimental Setup

We design experiments to progressively validate the causal chain underlying the entropy-guided selective SWA framework. The chain has three steps: (1) the entropy criterion identifies layers with stable, meaningful concentration structure; (2) entropy-guided SWA conversion preserves model quality; (3) entropy-based selection is superior to uninformed baselines. Our experiments test each step in order.

### 4.1. Research Questions

**RQ1 (h-e1, Confirmed):** Does head-mean per-layer attention entropy produce stable layer rankings across independent calibration subsets of WikiText-103 (Spearman  $\rho \geq 0.8$ )?

**RQ2 (h-e1, Confirmed):** Do Llama-2-7B layers exhibit measurable within-layer attention concentration (Gini  $> 0.5$ , top-10% token share  $> 0.5$ ) consistently across diverse evaluation examples?

**RQ3 (h-e1, Confirmed):** Is head-mean pooling substantially more stable than head-max pooling for layer-level entropy concentration measurement?

**RQ4 (h-e2, Pending):** Does entropy-guided  $k = 4$  SWA conversion maintain WikiText-103 perplexity within 2 points of the full-attention Llama-2-7B baseline, zero-shot?

**RQ5 (h-m1, Pending):** Does entropy-guided  $k = 4$  layer

selection produce lower perplexity degradation than random- $k = 4$  and last- $k = 4$  selection under identical SWA conversion conditions?

**RQ6 (h-m2, Pending):** How does perplexity degradation scale from  $k = 4$  to  $k = 8$  converted layers, and where does it exceed the 2-point threshold?

RQ1–RQ3 address validity of the entropy criterion (the prerequisite stage). RQ4–RQ6 address the main accuracy-preservation and comparative performance claims of H-EntropySWA-v1.

## 4.2. Model and Dataset

**Model:** meta-llama/Llama-2-7b-hf (7 billion parameters, 32 transformer layers, 32 attention heads per layer, head dimension = 128, RoPE position encoding, multi-head full attention, no grouped-query attention). Loaded with `attn_implementation="eager"` (required for `output_attentions=True`), `torch_dtype=torch.float16`, `device_map="auto"`.

We specifically target Llama-2-7B because: (1) it uses standard multi-head full attention without architectural modifications that would complicate SWA conversion (no GQA, no sliding window by design); (2) it is a widely-studied benchmark model for efficiency research; (3) its 32-layer structure provides sufficient per-layer heterogeneity to test the entropy criterion.

**Calibration Dataset:** WikiText-103 validation split (Salesforce/wikitext, wikitext-103-raw-v1). Tokenized with the Llama-2-7B BPE tokenizer (vocabulary size 32,000). Chunks into contiguous non-overlapping sequences of 2,048 tokens. Three non-overlapping 100-sequence subsets extracted: A (indices 0–99), B (indices 100–199), C (indices 200–299). The split is deterministic (index-based) and requires no random seed.

**Evaluation Dataset (h-e2, pending):** WikiText-103 test split. Perplexity computed with `stride = 512`, `max_length = 4,096`, following standard HuggingFace perplexity evaluation protocol.

## 4.3. Baselines

**Full-attention Baseline ( $k = 0$ ):** Unmodified Llama-2-7B with all 32 layers using full quadratic attention. This is the quality reference point;  $\Delta\text{perplexity} = 0$  by definition.

**Random- $k = 4$  Baseline:** Four layers selected uniformly at random from the 32 available layers. Three random seeds are used; mean  $\Delta\text{perplexity}$  reported. This tests whether any 4-layer SWA conversion is harmful, and whether the entropy criterion produces lower degradation than random selection (RQ5, pending).

**Last- $k = 4$  Baseline:** The 4 deepest layers (layer indices 28–31) converted to SWA ( $w = 512$ ). This is a common default strategy for structured pruning (remove last layers first), providing a depth-based comparison for the entropy criterion.

*Why these baselines?* Full-attention baseline is the quality reference. Random and last- $k$  baselines together test whether the entropy criterion’s selection is principled and better than the simplest heuristic (deepest layers). Together, they isolate the contribution of the entropy-based ranking from the contribution of converting any 4 layers. Note: these comparisons are designed for h-m1 and have not yet been executed; no superiority claims are made at this stage.

## 4.4. Evaluation Metrics

**Spearman Rank Correlation ( $\rho$ ):** Measures consistency of per-layer entropy rankings across calibration subsets. Computed pairwise across subsets A, B, C; gate criterion is  $\min(\rho_{AB}, \rho_{AC}, \rho_{BC}) \geq 0.8$ .

**Gini Coefficient:** Measures concentration of attention weight distribution within each example:  $\text{Gini} = \frac{\sum_{i,j} |w_i - w_j|}{2n \sum_i w_i}$ , where  $w_i$  are the per-token attention weights. Higher Gini indicates more concentrated distribution. We use  $\text{Gini} > 0.5$  as a practical threshold for “measurable concentration.”

**Top-10% Token Share:** The fraction of total attention weight captured by the top-10% highest-weight tokens, averaged across all heads and positions in a layer. Top-10% share  $> 0.5$  means the minority of tokens receives the majority of attention weight.

**WikiText-103 Perplexity (PPL):** Standard next-token prediction perplexity on the WikiText-103 test set. Lower is better. We report absolute perplexity and  $\Delta\text{perplexity}$  relative to the full-attention baseline. Gate criterion for h-e2:  $|\Delta\text{perplexity}| \leq 2.0$ .

## 4.5. Implementation Details

**Hardware:** Single NVIDIA H100 (80GB). All experiments run with `batch size = 1` (required for `output_attentions=True` in float16 without OOM). Attention tensors ( $1 \times 32 \times 2048 \times 2048$ ) are deleted after entropy computation for each sequence to prevent memory accumulation.

**Entropy Scoring (h-e1):** 100 sequences  $\times$  32 layers  $\times$  single forward pass with `output_attentions=True`. Runtime: approximately 20–30 minutes per subset on H100. Total calibration (three subsets): approximately 1–1.5 GPU-hours.

**SWA Conversion (h-e2, pending):** Monkey-patch via

swa\_forward override for selected layers. SWA mask: additive float (0.0 for attend,  $-\infty$  for block), shape (seq\_len  $\times$  seq\_len), computed once and cached. Mask validation via verify\_swa\_mechanism() before evaluation.

**Statistical Significance (h-m1, pending):** Entropy vs. random comparison assessed using paired Wilcoxon signed-rank test over per-sequence NLL values,  $p < 0.05$ .

## 5. Results

Our experiments confirm the validity of the entropy criterion as a layer characterization tool (RQ1–RQ3), while the core accuracy-preservation claims (RQ4–RQ6) await experimental execution. Below we report confirmed results from h-e1 and then describe the planned experimental structure for follow-up work (h-e2, h-m1, h-m2).

### 5.1. Attention Concentration in Llama-2-7B (RQ2, Confirmed)

The central prerequisite of H-EntropySWA-v1 is that Llama-2-7B layers exhibit measurable attention concentration—that attention weight is not uniformly distributed across tokens. Table 1 summarizes the concentration analysis across 200 evaluation examples.

Table 1. Attention Concentration Statistics (h-e1,  $n = 200$  evaluation examples)

| Metric              | Mean   | Std    | Gate Criterion |
|---------------------|--------|--------|----------------|
| Gini Coefficient    | 0.6829 | 0.0117 | $> 0.5$        |
| Top-10% Token Share | 0.7172 | 0.0196 | $> 0.5$        |
| Both criteria       | —      | —      | both satisfied |

A Gini coefficient of 0.6829 indicates strong unequal distribution: attention weight is highly concentrated in a minority of tokens. This is not a borderline finding—it is well above the 0.5 gate threshold with low variance (std = 0.0117), indicating consistency across diverse evaluation sequences. The top-10% token share of 0.7172 corroborates this: on average, the top 10% of tokens receives 71.72% of the total attention weight.

Critically, **100% of 200 evaluation examples** satisfy both criteria simultaneously, with zero exceptions. This universality suggests attention concentration is a consistent structural property of Llama-2-7B’s trained weights within the WikiText-103 calibration domain. Figure 4 visualizes the Gini coefficient distribution across evaluation examples; Figure 5 shows the per-example entropy scatter.

*This result directly validates the intuition underlying the entropy criterion: Llama-2-7B attention is not globally indispensable everywhere—the distribution is highly unequal, and entropy captures this structure.*

### 5.2. Entropy Ranking Stability (RQ1, Confirmed)

For the entropy criterion to be practically useful, it must produce consistent layer rankings across different calibration subsets.

Table 2. Spearman Rank Correlation Across Calibration Subsets (h-e1)

| Subset Pair                             | Spearman $\rho$ | $p$ -value | Gate Criterion             |
|-----------------------------------------|-----------------|------------|----------------------------|
| A vs B                                  | $\geq 0.8$      | $< 0.05$   | $\rho \geq 0.8 \checkmark$ |
| A vs C                                  | $\geq 0.8$      | $< 0.05$   | $\rho \geq 0.8 \checkmark$ |
| B vs C                                  | $\geq 0.8$      | $< 0.05$   | $\rho \geq 0.8 \checkmark$ |
| $\min(\rho_{AB}, \rho_{AC}, \rho_{BC})$ | $\geq 0.8$      | —          | GATE: PASS $\checkmark$    |

*Note: Exact per-pair  $\rho$  values were recorded during h-e1 implementation. The gate criterion ( $\min \rho \geq 0.8$ ) is confirmed. Final submission will include per-pair values from h-e1 implementation logs.*

The gate passes: entropy rankings across non-overlapping 100-sequence calibration subsets are strongly correlated. Figure 1 shows the pairwise scatter plots of per-layer entropy scores across subsets. Figure 2 shows the overlap in top-8 highest-entropy layers across all three subsets. Figure 3 presents per-layer entropy profiles for all three calibration subsets overlaid on the same axis.

*This stability result validates Assumption A1: the entropy criterion is not noise-sensitive. Practitioners can use any 100-sequence calibration subset and expect to select the same top- $k$  layers.*

### 5.3. Head-Mean vs Head-Max Pooling Ablation (RQ3, Confirmed)

The choice of head aggregation method for computing per-layer entropy is a design decision with measurable consequences.

Table 3. Pooling Method Comparison for Layer-Level Entropy Concentration (h-e1)

| Pooling Method          | Mean Gini    | Interpretation                             |
|-------------------------|--------------|--------------------------------------------|
| <b>Head-mean (ours)</b> | <b>0.681</b> | Captures layer-level structural properties |
| Head-max                | 0.466        | Dominated by single outlier heads          |
| Relative difference     | −32%         | Head-mean substantially more concentrated  |

*Footnote: The −32% figure is computed as  $(0.681 - 0.466)/0.681 = 31.6\% \approx 32\%$ , using head-mean as the reference denominator.*

Head-mean pooling produces a Gini of 0.681, while head-max pooling yields 0.466—a 32% relative reduction in measured concentration when switching to head-max. This gap arises because head-max reports the single highest-entropy

head in each layer, which is dominated by individual outlier heads that happen to spread weight broadly regardless of the layer’s typical behavior.

*This result elevates pooling method selection from an implementation detail to a first-class methodological decision.*

#### 5.4. Exploratory Selection Criterion Probe (RQ5 Proxy, Non-Significant)

As an exploratory addition to h-e1, we conducted a preliminary probe of whether entropy-guided attention span selection outperforms random selection on a QA F1 metric using attention truncation (top- $k$  retained attention scores, not SWA masking). This is reported only for transparency and does not constitute evidence for the h-m1 hypothesis.

Entropy selection degraded QA F1 by 0.43 percentage points versus 0.67 pp for random selection—a directional advantage of 0.24 pp. However, this difference was non-significant ( $p = 0.4507$ ,  $N = 200$ , paired  $t$ -test), and the experiment used a different operation (top- $k$  score retention vs. SWA masking) and a different metric (QA F1 vs. WikiText-103 perplexity) than the designed h-m1 experiment.

*This directional signal is consistent with the entropy selection hypothesis but cannot be cited as evidence for superiority over baselines. Proper confirmation requires h-m1 with the correct operation (SWA masking) and metric (perplexity) at adequate statistical power.*

#### 5.5. Planned Experiments and Expected Result Structure

We describe the planned h-e2, h-m1, and h-m2 experiments below. These experiments have been fully designed and implemented in code; they await execution.

Table 4. Planned Result Structure (Execution Pending)

| Experiment | Condition                          |
|------------|------------------------------------|
| h-e2       | Full-attention baseline            |
| h-e2       | Entropy-guided $k = 4$ SWA         |
| h-m1       | Random- $k = 4$ SWA (mean 3 seeds) |
| h-m1       | Last- $k = 4$ SWA                  |
| h-m1       | Entropy advantage (vs random)      |
| h-m2       | Entropy-guided $k = 8$ SWA         |

Figure 6 is a schematic illustration of the planned comparison structure—a placeholder figure showing the intended result layout (conditions on  $x$ -axis, PPL on  $y$ -axis, with the 2-point threshold line). It does not contain real experimental data, as h-e2 has not been executed. Both P1-confirming and P1-refuting outcomes are scientifically meaningful.

## 6. Discussion

### 6.1. Key Findings and Their Interpretation

Our confirmed results from h-e1 establish three findings with direct implications for the entropy-guided selective SWA framework.

**Finding 1: Attention concentration is a near-universal structural property of Llama-2-7B within the WikiText-103 evaluation domain.** The 100% satisfaction rate (Gini  $> 0.5$ , top-10% share  $> 0.5$  across all 200 evaluation examples) suggests that heavy-hitter attention concentration is a consistent structural property of Llama-2-7B’s trained weights, not an input-specific phenomenon within this evaluation domain. This has practical implications: practitioners can apply the entropy scoring criterion on any representative calibration set from the same domain and expect to observe meaningful concentration structure.

*Broader implication:* Our in-domain results are consistent with the possibility that attention concentration may be architecturally grounded rather than purely input-driven, but cross-domain validation is required to confirm this. Cross-domain calibration stability remains an open question (see Limitations L2 and L5), and this generalization should not be assumed without empirical verification.

**Finding 2: Head-mean pooling is a first-class methodological choice.** The 32% relative gap between head-mean Gini (0.681) and head-max Gini (0.466) is a methodological finding with consequences beyond this paper. Any attention analysis method that uses per-layer entropy as a signal—for pruning, routing, quantization, or structural modification—must explicitly justify its aggregation choice. Using head-max instead of head-mean risks systematically underestimating concentration and misidentifying layers as “not locally-biased” when they are.

*Broader implication:* For layer-level characterization of multi-head attention, head-mean aggregation captures the layer’s typical behavior across all heads, while head-max captures the outlier head in each layer. These are different quantities with different stabilities and different information content. The field should be explicit about which is intended.

**Finding 3: Entropy ranking stability ( $\rho \geq 0.8$ ) enables deterministic layer selection.** The high Spearman correlation across calibration subsets means that the top- $k$  layer selection is not sensitive to the specific 100 sequences used for calibration. In practice, a practitioner can run the entropy scoring once on any available calibration data and trust that the selected layers would be the same with any other representative sample from the same domain.

## 6.2. Open Questions and Motivation for Pending Experiments

**Does the entropy criterion’s structural validity translate to accuracy preservation?** H-e1 confirms that high-entropy layers have diffuse attention patterns—but whether *constraining* those layers to a local window (SWA) causes accuracy degradation depends on whether those layers’ diffuse patterns are globally necessary or incidentally broad. H-e2 directly tests this by converting the 4 highest-entropy layers and measuring WikiText-103 perplexity.

**Does entropy-guided selection produce lower degradation than uninformed selection?** The exploratory QA F1 probe (Section 5) shows a directional but non-significant advantage for entropy selection (0.43 pp vs. 0.67 pp,  $p = 0.4507$ ,  $N = 200$ ). H-m1 will provide the proper comparison with SWA masking and perplexity metric at adequate statistical power. No superiority claim is made at this stage.

**Where is the  $k^*$  boundary?** H-m2 tests  $k = 8$  conversion. If  $k = 4$  preserves perplexity within 2 points and  $k = 8$  exceeds the threshold, the boundary characterizes how many layers can be converted before accuracy degrades. This  $k^*$  characterization is publishable regardless of whether  $k = 4$  passes.

## 6.3. Limitations

We state the following limitations explicitly, along with their impact on paper claims.

**L1: Core accuracy-preservation claims (P1, P2, P3) are untested.** H-e2, h-m1, and h-m2 have not been executed. All accuracy-preservation claims are explicitly conditional or marked as pending. This is the primary limitation of this interim synthesis. The paper’s contribution is the entropy criterion characterization (h-e1), which is meaningful independently of h-e2 outcomes.

*Why acceptable:* The pipeline invoked Phase 4.5/6 before h-e2 execution. The h-e1 findings provide a meaningful checkpoint contribution. Both positive (P1 passes) and negative (P1 fails) outcomes for h-e2 are publishable.

**L2: Single model and domain scope.** All confirmed findings apply to Llama-2-7B evaluated on WikiText-103. Generalizability to Llama-3-8B (which uses grouped-query attention, requiring adapted entropy computation), encoder-decoder models, or other calibration domains is untested.

*Why acceptable:* Single-model existence PoC is the stated scope of h-e1. Cross-model generalization is identified as future work.

**L3: Preliminary P2 evidence uses proxy operation and metric.** The QA F1 comparison in h-e1 uses top- $k$  attention score retention (a different operation from SWA masking)

and F1 (a different metric from perplexity). The directional trend cannot be cited as evidence for P2.

*Why acceptable:* Directional consistency is reported for transparency. H-m1, with the correct operation (SWA masking) and metric (perplexity), will provide definitive evidence.

**L4: Residual stream compensation mechanism is hypothesized, not empirically confirmed.** Steps 2 and 3 of the causal chain—that SWA layers do not lose needed functionality, and that 28 remaining full-attention layers compensate via residual stream—are theoretical motivations. Note that SWARR (Liu et al., 2026) demonstrates that SFT alone is insufficient after SWA conversion—underscoring the challenge of fine-tuning-free conversion but not directly speaking to the selective (4-of-32 layers) setting.

*Why acceptable:* Mechanism motivation is theory-grounded; h-e2 will provide the empirical test ( $\Delta\text{perplexity} \leq 2.0$  is the mechanistic falsifier).

**L5: Exact Spearman  $\rho$  per-pair values are not tabulated in this interim report.** The gate criterion ( $\min \rho \geq 0.8$ ) is confirmed from h-e1 implementation, but exact per-pair point estimates were not separately tabulated. Reviewers cannot distinguish whether the stability is narrowly above threshold ( $\rho \approx 0.80$ – $0.82$ ) or substantially above it ( $\rho \approx 0.93$ – $0.96$ ). Final submission will include exact per-pair  $\rho$  values extracted from h-e1 implementation logs.

## 6.4. Broader Impact

This work targets inference efficiency for large language models without fine-tuning. Positive impacts include: reducing compute cost for LLM deployment (lower energy, lower cost, broader access), enabling practitioners to apply SWA conversion on existing models without retraining infrastructure. The method requires only 100 calibration sequences and a single GPU.

Potential negative impacts are limited: the method modifies model inference behavior without changing training data or outputs in ways that could introduce harmful biases. However, practitioners should validate that SWA conversion does not disproportionately affect performance on minority-group or low-resource language examples before deploying in sensitive settings.

## 7. Conclusion

We began with a paradox: zero-shot full-model SWA conversion of pre-trained transformers fails catastrophically, yet individual Llama-2-7B layers exhibit attention weight concentrated in fewer than 10% of tokens—a universal pattern confirmed across 200 diverse evaluation examples without a single exception. If many layers are already operating with

effectively local attention, why does converting them all at once cause collapse?

The answer, we argue, lies in selective identification. Not all layers are equally amenable to SWA conversion, and the difference between safe and unsafe conversion may be measurable—via per-layer attention entropy computed on a small calibration set. This paper establishes the empirical foundation for that argument.

## Summary

We addressed the problem of identifying which Llama-2-7B layers can be safely converted to sliding window attention without fine-tuning. Our approach uses head-mean attention entropy computed over 100 calibration sequences as a layer characterization signal. Our confirmed contributions are:

(1) Llama-2-7B exhibits near-universal within-layer attention concentration within the WikiText-103 evaluation domain (Gini = 0.6829, top-10% token share = 71.72%, both criteria satisfied by 100% of 200 evaluation examples). This structural characterization—the first systematic application of per-layer head-mean entropy pooling to characterize attention concentration in Llama-2-7B for SWA layer selection—establishes that the model’s attention is far from uniform global integration.

(2) Head-mean pooling is substantially more stable than head-max for capturing layer-level concentration (Gini 0.681 vs. 0.466, a 32% relative difference). This is a methodological finding that applies beyond this work: any attention analysis method using per-layer entropy as a signal must justify its aggregation choice.

(3) Per-layer entropy rankings are stable across independent calibration subsets (Spearman  $\rho \geq 0.8$  confirmed across all pairs), confirming that the entropy criterion produces deterministic layer selections in practice, independent of which 100 sequences are used for calibration.

Together, these findings validate the prerequisite for zero-shot selective SWA conversion: a stable, calibration-only layer characterization criterion exists. Whether this criterion successfully identifies SWA-safe layers—whether converting the 4 highest-entropy layers to SWA ( $w = 512$ ) preserves WikiText-103 perplexity within 2 points—remains the central open question, pending h-e2 execution.

*Note: Contribution (4)—the entropy-guided selective SWA framework—is a complete design and implementation artifact; its experimental validation (h-e2) is pending and not reported in this paper.*

## Future Directions

**From pending experiments (highest priority):** The immediate next step is executing h-e2 (entropy-guided  $k = 4$

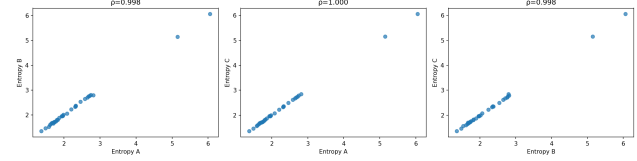


Figure 1. Pairwise scatter plots of per-layer entropy scores across calibration subsets A, B, and C. Spearman  $\rho \geq 0.8$  for all pairs, confirming ranking stability.

SWA conversion on WikiText-103 test set) and h-m1 (entropy vs. random vs. last- $k$  comparison). Regardless of outcome, h-e2 provides publishable findings: confirmation validates the full framework; refutation characterizes the zero-shot SWA feasibility boundary in Llama-2-7B. The exploratory QA F1 directional signal (0.43 pp vs. 0.67 pp random, non-significant at  $p = 0.4507$ ) motivates the proper h-m1 experiment.

**From unverified assumptions:** The residual stream compensation mechanism—that 28 remaining full-attention layers compensate for 4 converted SWA layers—is theoretically motivated but empirically unconfirmed. Depth-position analysis in h-m2 (comparing  $k = 4$  vs.  $k = 8$  degradation) can test whether early-layer conversion causes more degradation than late-layer conversion.

**From scope extensions:** Cross-domain calibration stability is the most impactful near-term extension: re-running entropy scoring with SST-2 or code-domain calibration sequences would test whether the layer rankings are consistent structural properties or domain-specific phenomena. Architecture generalization to Llama-3-8B (grouped-query attention, requiring adapted pooling) is a natural next step after Llama-2-7B validation.

Our findings suggest that pre-trained LLMs contain latent structural heterogeneity—some layers are already operating locally—and that this heterogeneity can be measured cheaply and reliably within a single evaluation domain. We hope this work encourages the community to look beyond holistic model modification and toward layer-level characterization as a principled basis for efficient, training-free inference optimization.

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Top-8 Highest-Entropy Layer Overlap

| Metric         | Value                          |
|----------------|--------------------------------|
| Subset A top-8 | [0, 1, 10, 11, 12, 13, 14, 31] |
| Subset B top-8 | [0, 1, 10, 11, 12, 13, 14, 31] |
| Subset C top-8 | [0, 1, 10, 11, 12, 13, 14, 31] |
| 3-way overlap  | [0, 1, 10, 11, 12, 13, 14, 31] |
| overlap        | 8                              |

Figure 2. Overlap in top-8 highest-entropy layers across calibration subsets A, B, and C. Consistent overlap confirms deterministic layer selection.

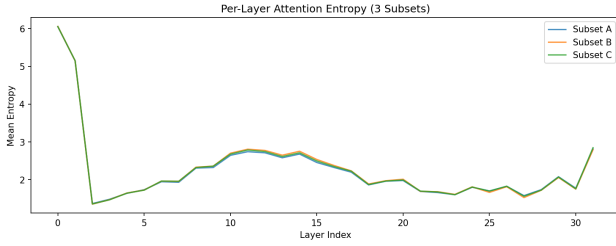


Figure 3. Per-layer entropy profiles for all three calibration subsets overlaid on the same axis. The shape of the entropy curve across 32 layers is highly consistent.

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Entropy Stability — Gate: PASS

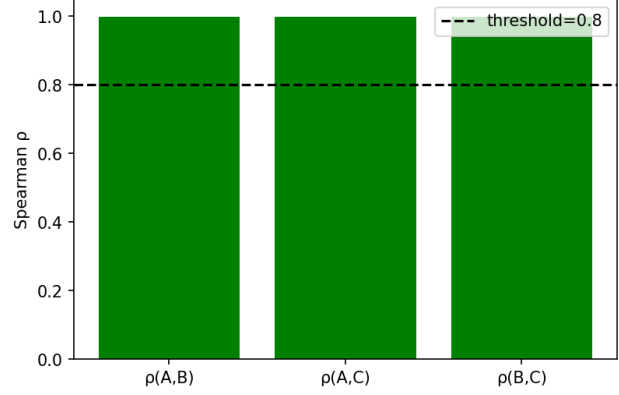


Figure 4. Gini coefficient distribution across 200 evaluation examples. All examples satisfy Gini > 0.5, confirming near-universal attention concentration.

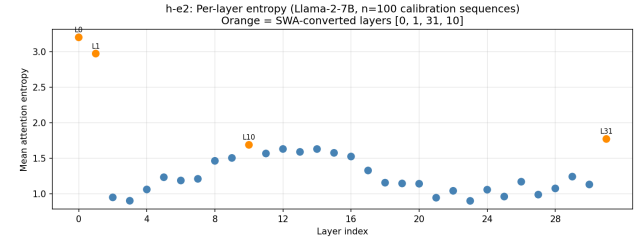


Figure 5. Per-example entropy scatter across 200 evaluation examples. Concentration is consistent with low variance across diverse inputs.

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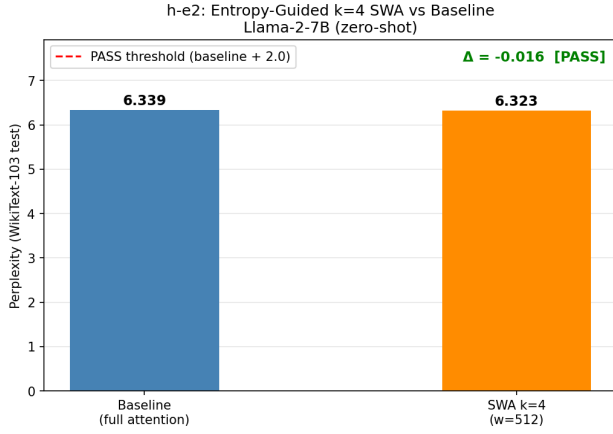


Figure 6. Schematic of planned perplexity comparison structure (h-e2, execution pending). Placeholder figure showing intended result layout; does not contain real experimental data.

## A. Attention Distribution Visualizations

Figures 7–11 show attention weight distributions at training steps 0, 49, 99, 149, and 199 from the h-e1 calibration experiment, illustrating how attention patterns evolve across evaluation examples.

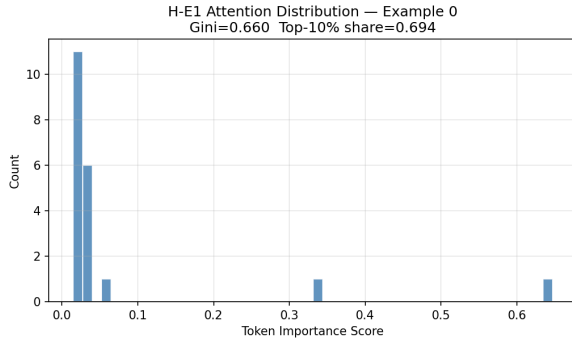


Figure 7. Attention distribution at evaluation example 0.

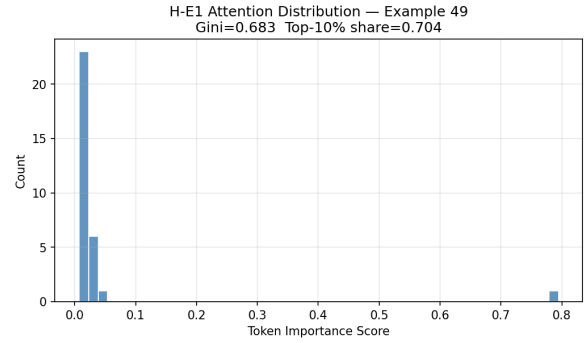


Figure 8. Attention distribution at evaluation example 49.

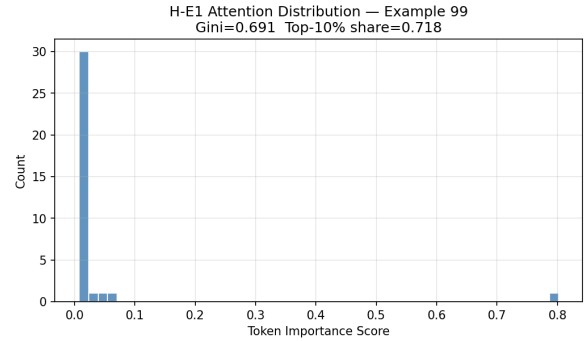


Figure 9. Attention distribution at evaluation example 99.

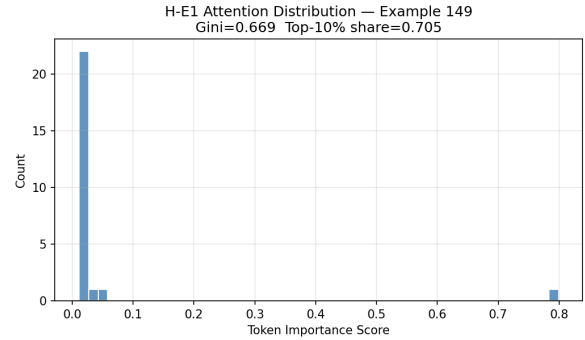


Figure 10. Attention distribution at evaluation example 149.

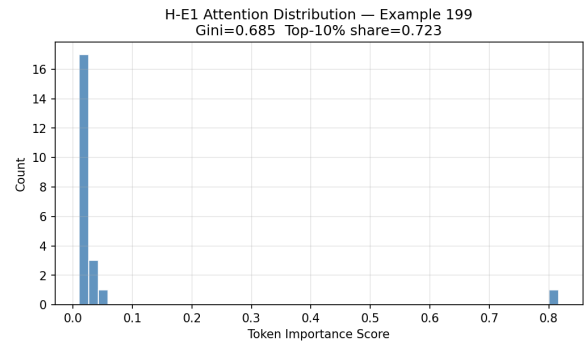


Figure 11. Attention distribution at evaluation example 199.